

pystackr: An R Package for Bayesian Plausible-Value Analysis with One Stacked Fit Calibrated to Replicate-Weight Variance Targets

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ABSTRACT. Large-scale assessments such as PISA report student proficiency as a set of plausible values and provide replicate weights for design-based variance estimation. Established R packages accommodate both conventions for frequentist analysis. For a Bayesian analysis, however, the combining convention requires one Markov chain Monte Carlo run per plausible value, and there has been no established way of connecting posterior uncertainty to the replicate weights. In this paper we describe the R package `pystackr`, which addresses both problems for the fixed effects of survey-weighted linear models. All plausible values enter a single stacked fit, run through a bundled `brms` backend or through any Bayesian engine the user attaches. A separate target stage refits the model deterministically by weighted least squares, once per plausible value and replicate weight, and pools the results into a balanced-repeated-replication covariance target by the multiple-imputation combining rules. An exact affine correction then calibrates the posterior draws so that their mean and covariance equal the target, and each reported row records whether its interval is coverage-claimable. The methodology is applied to reading proficiency in the American sample of PISA 2022, where within-school and between-school gradients of socioeconomic status, gender, and immigrant background are estimated from one sampler run and reported with posterior probability statements that carry design-based uncertainty. The package is open source and documented with ten vignettes.

Keywords: plausible values; replicate weights; balanced repeated replication; Bayesian inference; multiple imputation; PISA; R

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1. Introduction

Secondary analyses of large-scale assessment data must accommodate two conventions of the reporting programs. First, programs such as PISA, TIMSS, and NAEP do not report a student’s proficiency as a single score. Because each student answers only a subset of the item pool, proficiency is a latent quantity, and the programs release a set of M plausible values (PVs), random draws from each student’s conditional proficiency distribution (Mislevy, 1991; Mislevy et al., 1992; von Davier et al., 2009). The convention is to treat the plausible values as multiple imputations: the analysis is repeated once per plausible value and the M results are combined with the rules of Rubin (1987). Second, students are selected by stratified multistage sampling, and standard errors are expected to reflect that design. The programs therefore distribute final weights together with sets of replicate weights that encode the stratification and clustering of the first-stage sample; PISA supplies 80 balanced-repeated-replication (BRR) weights with Fay’s modification (Judkins, 1990; OECD, 2024). Both conventions are well supported in R for frequentist estimation, for instance by EdSurvey (Zhang et al., 2024), RALSA (Mirazchiyski, 2021), intsvy (Caro & Biecek, 2017), Rrepest (Ilizaliturri et al., 2026), and BIFIEsurvey (BIFIE et al., 2026), and applied secondary analyses routinely depend on both (e.g. Lee & Kaur, 2026).

Bayesian estimation is increasingly advocated for the analysis and reporting of large-scale assessments (Kaplan, 2016; Kaplan & Harra, 2024). A posterior distribution supports summaries that a significance test does not provide, such as the probability that an effect lies between zero and its estimated value, and Kaplan and Harra (2024) organize a nine-step workflow for assessment data around summaries of this kind. General-purpose samplers such as Stan (Carpenter et al., 2017) and brms (Bürkner, 2017) have made survey-weighted regression models straightforward to fit. Two obstacles remain, however. The first is computational: under the combining convention a Bayesian analysis requires one Markov chain Monte Carlo (MCMC) run per plausible value, each with its own convergence assessment, before the draws can be mixed across imputations (Zhou & Reiter, 2010). The second obstacle concerns the variance of the reported estimates. Posterior spread reflects the likelihood and the prior rather than the sample design, and the replicate-weight machinery that defines design-based uncertainty in official reporting has had no Bayesian counterpart. Kaplan and Harra (2024) acknowledge this gap explicitly: they did not use BRR weights because no research had examined them in a Bayesian context, and they leave the problem to future work. Similarly, Zheng et al. (2026) show that replication methods

remain valid for two-level models and call for expanded software support; as far as we are aware, no available software addresses the Bayesian case.

In this paper we describe the R package `pvstackr`, which addresses both obstacles for the fixed effects of survey-weighted linear models. All M plausible values enter one stacked fit, which is run once, through a bundled `brms` backend or through any Bayesian engine the user attaches. A separate target stage refits the same model deterministically by weighted least squares, once per plausible value and replicate weight, and assembles the results into a design-based fixed-effect covariance target, the BRR-Fay sandwich within each plausible value pooled across plausible values by the combining rules. An exact affine correction then calibrates the stacked posterior draws so that their mean and covariance equal the target. The analyst therefore retains the posterior draws, and with them probability statements and derived quantities, while the reported point estimates, standard errors, and degrees of freedom are the design-based ones. Each output row carries metadata recording whether its interval is coverage-claimable and why. The statistical theory, including the point identity for the stacked fit and simulation evidence of near-nominal interval coverage, is developed in a companion paper (Lee et al., 2026); the present article concerns the software.

We make no claim that one stacked fit is faster than M fits of the same sampler; the stacked data set has nM rows, and sampling cost scales accordingly. The computational advantage lies instead in the amount of stochastic computation: a single fit entails a single model object and a single convergence assessment instead of M , and the design-based variance is obtained from $M(R + 1)$ least-squares refits rather than from $M(R + 1)$ MCMC runs. In the application below the target stage takes well under one second, whereas the corresponding brute-force Bayesian computation, one MCMC run per plausible value and replicate weight, would require 810 sampler runs.

1.1. Competing software. Table 1 places `pvstackr` among the tools an applied analyst might consider. The design-based packages mentioned above automate plausible-value combining and replicate-weight variance estimation for descriptive statistics and regression, and `EdSurvey` additionally fits weighted mixed models through `WeMix` (Bailey et al., 2023); none of them produces a posterior distribution. General Bayesian engines produce posterior distributions but leave plausible-value combining and design-based variance to the user, which in practice means M separate runs and model-based intervals. `pvstackr` combines the two approaches, producing a single Bayesian fit whose reported uncertainty is the design-based target while the

TABLE 1. Feature comparison with related R software

	EdSurvey	RALSA	intsvy	BIFIEsurvey	brms	pvstackr
Plausible-value combining	✓	✓	✓	✓	manual	✓
Replicate-weight standard errors	✓	✓	✓	✓	—	✓
Posterior draws for reporting	—	—	—	—	✓	✓
Design-calibrated posterior	—	—	—	—	—	✓
Coverage metadata per row	—	—	—	—	—	✓
Model fits per analysis	$M(R+1)$	$M(R+1)$	$M(R+1)$	$M(R+1)$	M MCMC	1 MCMC ¹

Note. M = number of plausible values; R = number of replicate weights. “Manual” means the workflow is possible but must be assembled by the user. **Rrepest** (Ilizaliturri et al., 2026) follows the same pattern as the first four columns; **WeMix** (Bailey et al., 2023) fits weighted mixed models and relies on **EdSurvey** for the plausible-value loop. Direct-estimation packages cover many analysis types beyond regression; **pvstackr** 0.2.0 covers fixed effects of weighted linear models.

¹Plus $M(R+1)$ weighted least-squares fits for the covariance target, which are deterministic and cost seconds.

posterior draws are retained. Its scope is narrower than that of the direct-estimation packages; version 0.2.0 covers fixed effects of weighted linear models.

The plan of the article is as follows. In the Background section we review the combining convention, replicate-weight variance estimation, and the calibrated stacked workflow that connects them. The ‘About the tool’ section describes the architecture, the interface, and the safeguards. In the Applications section the workflow is applied to reading proficiency in the PISA 2022 American sample and compared with the conventional per-plausible-value route. The final section presents conclusions and planned development.

2. Background

2.1. Plausible values and the combining convention. Plausible values are draws from each student’s conditional distribution of latent proficiency given the student’s item responses and background variables, generated by the assessment’s latent regression model (Jewsbury et al., 2025; Mislevy et al., 1992). A single plausible value behaves like an error-prone observation of proficiency, whereas the set of M values carries the measurement uncertainty that is lost when a single score is analyzed (Rutkowski et al., 2010; von Davier et al., 2009). The set is analyzed by the multiple-imputation convention (Meng, 1994; Rubin, 1996). If $\hat{\beta}_m$ and $\hat{U}_m$ denote the point estimate and its covariance from the analysis of plausible value m , the combined

estimate and total covariance are

$$\bar{\beta}_{\text{MI}} = \frac{1}{M} \sum_{m=1}^M \hat{\beta}_m, \quad T_{\text{MI}} = \bar{U} + \left(1 + \frac{1}{M}\right) B, \quad (1)$$

where $\bar{U} = M^{-1} \sum_m \hat{U}_m$ is the average within-imputation covariance and $B = (M - 1)^{-1} \sum_m (\hat{\beta}_m - \bar{\beta}_{\text{MI}})(\hat{\beta}_m - \bar{\beta}_{\text{MI}})^\top$ is the between-imputation covariance (Rubin, 1987). Intervals use a t reference distribution; the small-sample degrees of freedom of Barnard and Rubin (1999) additionally require the complete-data degrees of freedom the analysis would have had with no missing information. The diagonal ratio $\gamma_k = (1 + 1/M)B_{kk}/T_{\text{MI},kk}$ is the fraction of missing information (FMI), the share of total variance attributable to proficiency measurement rather than sampling (Cortes et al., 2025). Analyses based on a single plausible value drop the B term entirely and understate uncertainty by exactly this share; the shortcut nevertheless remains common in applied work (Jewsbury et al., 2024; Laukaityte & Wiberg, 2017).

For a Bayesian analysis the same convention applies with posterior draws in place of point estimates: the model is fitted once per imputation and the draws are mixed across imputations (Zhou & Reiter, 2010). Mixing is statistically straightforward but computationally linear in M , and each of the M MCMC runs requires its own convergence assessment before its draws can be used (Kaplan & Harra, 2024).

2.2. Design-based variance and replicate weights. Large-scale assessments sample schools first, with stratification, and then students within schools, so observations are neither independent nor equally weighted (Mang et al., 2021; Rutkowski et al., 2010). Design-based variance estimation proceeds by recomputing the point estimate under systematic perturbations of the first-stage units. In the BRR scheme with Fay’s modification, schools are paired into R variance strata; replicate r multiplies the weights of one half-sample by $2 - k$ and the other by k , with $k = 0.5$ in PISA, so that no observation is ever fully dropped (Judkins, 1990; OECD, 2024). If $\hat{\beta}_m^{(r)}$ is the estimate for plausible value m under replicate weight r , the design-based covariance for that plausible value is

$$\hat{U}_m^{\text{BRR}} = a_d \sum_{r=1}^R (\hat{\beta}_m^{(r)} - \hat{\beta}_m)(\hat{\beta}_m^{(r)} - \hat{\beta}_m)^\top, \quad a_d = \frac{1}{R(1 - k)^2}, \quad (2)$$

which is $a_d = 0.05$ under the PISA constants $R = 80$ and $k = 0.5$. Substituting $\hat{U}_m^{\text{BRR}}$ for $\hat{U}_m$ in Equation (1) yields a total covariance whose sampling component is design-based and whose measurement component is the usual between-imputation

term. The replicate weights encode the stratification and clustering that produced the data, so the analyst does not need to reconstruct variance strata by hand (Zheng et al., 2026); model-based standard errors from the same weighted regression, by contrast, reflect neither (Rutkowski et al., 2010), and weighting choices alone change interval coverage substantially (Atasever et al., 2025).

Note that a replication loop estimates the variance of whatever quantity is recomputed per replicate. Applied to the analysis of a single plausible value, it recovers the within component $\hat{U}_m^{\text{BRR}}$ only; the between-imputation component B is defined across analyses of different plausible values and must still come from Equation (1) (Lee et al., 2026). The two variance components are thus obtained by different means, replication for the sampling part and the combining rules for the measurement part, and software that automates only one of them leaves the uncertainty statement incomplete.

2.3. One stacked fit, calibrated to the target. `pvstackr` implements a workflow, developed in Lee et al. (2026), that reduces the M MCMC runs to one. The M plausible-value columns are stacked into nM rows, and each row enters the likelihood at weight $\tilde{w}_i/M$, where $\tilde{w}_i$ is the final student weight normalized to mean one. The resulting posterior is the stacked fractional posterior

$$q(\boldsymbol{\beta}, \sigma) \propto p(\boldsymbol{\beta}, \sigma) \prod_{m=1}^M \tilde{L}_m(\boldsymbol{\beta}, \sigma)^{1/M}, \quad (3)$$

where $\tilde{L}_m$ is the weighted likelihood under plausible value m , with the final weights entering as a survey pseudolikelihood (Pfeffermann et al., 1998; Rabe-Hesketh & Skrondal, 2006). The $1/M$ exponent makes the data enter as a geometric average of the M per-imputation likelihoods, so the stacked posterior carries one sample’s worth of information rather than M ; the stacking is fractional pooling of imputations within one fit, not the posterior model stacking of Yao et al. (2018). Under a flat prior on the coefficients, the maximizer of the stacked objective equals the average of the per-imputation weighted least-squares estimates exactly, which is the combining-rule center $\bar{\boldsymbol{\beta}}_{\text{MI}}$ of Equation (1) (Lee et al., 2026, Theorem 1). The stacked fit therefore reproduces the combining-rule center. Its spread, however, estimates neither the Rubin total T_{MI} nor any design-based quantity, and no single posterior can, because B is defined across separate per-imputation analyses (Lee et al., 2026).

The package therefore calibrates the stacked posterior. The target $(\bar{\boldsymbol{\beta}}_{\text{MI}}, T_{\text{MI}})$ is assembled outside the sampler from the $M(R+1)$ deterministic weighted least-squares fits of Equations (1) and (2). The calibration is the Cholesky calibration correction

(CCC): with L_t and L_{raw} denoting the lower-triangular Cholesky factors of T_{MI} and of the raw draw covariance Σ_{raw} , each fixed-effect draw β_s is mapped to

$$\beta_s^{\text{cal}} = \bar{\beta}_{\text{MI}} + A(\beta_s - \bar{\beta}_{\text{raw}}), \quad A = L_t L_{\text{raw}}^{-1}, \quad (4)$$

an exact affine transformation that is applied once and requires neither iteration nor tuning. The calibrated draws have mean $\bar{\beta}_{\text{MI}}$ and covariance T_{MI} to machine precision, and they retain the joint dependence structure of the stacked posterior beyond its first two moments. The rationale is that of calibrated Bayes, under which posterior summaries are useful to the extent that they carry frequentist guarantees under the design (Little, 2011); the transformation is the fixed-effect analogue of the scale and shape corrections that have been used for pseudo-posteriors under informative sampling (Savitsky & Williams, 2022; Williams & Savitsky, 2020, 2021). The coverage properties derive from the construction of the target rather than from the transformation itself: an interval is coverage-claimable when its covariance is the design-based total of Equations (1) and (2) and its reference distribution uses the Barnard and Rubin (1999) degrees of freedom. The package records exactly this condition for each row, as we describe in the next section. Version 0.2.0 restricts the workflow to fixed effects of survey-weighted linear models; variance components are passed through uncalibrated and are excluded from the reportable output.

3. About the tool

3.1. Architecture. `pystackr` is organized as five stages, four of them corresponding to a single exported function each and the fifth to a family of accessor functions, summarized in Table 2 and Figure 1. The package has no hard runtime dependencies beyond base R; the target stage runs on a built-in weighted least-squares engine, so the $M(R+1)$ replicate fits require no modeling library. For the sampler, the fit stage provides a bundled `brms` backend (Bürkner, 2017), selected by a single control argument; the backend is availability-guarded, so the package installs and runs without `brms` present, and Stan is called through `cmdstanr` (Gabry et al., 2024) when it is installed. Any other engine with weighted-likelihood support can be attached instead through three user functions, one fitting the prepared stacked model, one returning a draw matrix, and one reporting the sampler diagnostics; the package supplies these functions with a prepared data frame and formula, all three are required for a reportable fit, and the supplement lists the `brms` backend, whose three functions the package exports, injected in this form as a template.

TABLE 2. The five stages of the `pvstackr` workflow

Stage	Function	Role
Design	<code>pv_design()</code>	Declare plausible-value columns, final weight, replicate weights, and Fay coefficient; record hashes for provenance.
Target	<code>pv_brr_target()</code>	Assemble the external fixed-effect target: $M(R + 1)$ weighted least-squares fits, BRR-Fay covariance per plausible value, Rubin combining, degrees of freedom.
Fit	<code>pv_fit()</code>	One stacked fit through the bundled brms backend or an injected engine, then the Cholesky calibration to the target (<code>method = "stack_direct"</code>); reference and diagnostic paths via <code>"per_pv"</code> and <code>"stack_psis"</code> .
Compare	<code>pv_compare_methods()</code>	Align finished fits, difference them against a reference, record agreement and timing.
Read	<code>get_estimates()</code> , <code>get_target()</code> , <code>get_draws()</code> , <code>get_diagnostics()</code>	Accessor family; estimates arrive as a data frame carrying interval metadata.

Two further methods share the fit stage. `method = "per_pv"` is the conventional reference: it accepts M per-plausible-value draw matrices, from fits the user runs, and pools them with Equation (1) using model-based within covariances. `method = "stack_psis"` is a diagnostic path that reweights the stacked draws toward each per-plausible-value posterior using Pareto-smoothed importance sampling (PSIS) weights supplied by the user's PSIS routine, for example `loo::psis()`, and reports the tail-shape diagnostic $\hat{k}$ per plausible value (Vehtari et al., 2024); its purpose is to check the stacked fit rather than to replace the calibration, and it blocks its own estimates when any $\hat{k}$ reaches 0.7.

3.2. Interval metadata and safeguards. Each estimate row carries two fields that record the inferential status of its interval. `interval_role` names the construction behind the interval, and `coverage_claim_allowed` is TRUE for exactly one construction: a `stack_direct` row calibrated to an external BRR-Fay target combined by Equation (1), with Barnard and Rubin (1999) degrees of freedom computed from a user-stated complete-data value. Classic Rubin degrees of freedom, per-plausible-value reference rows, and importance-sampling rows are all labeled descriptive. Because the label is part of the output table, a reader of derived results can always establish what kind of uncertainty statement each interval represents.

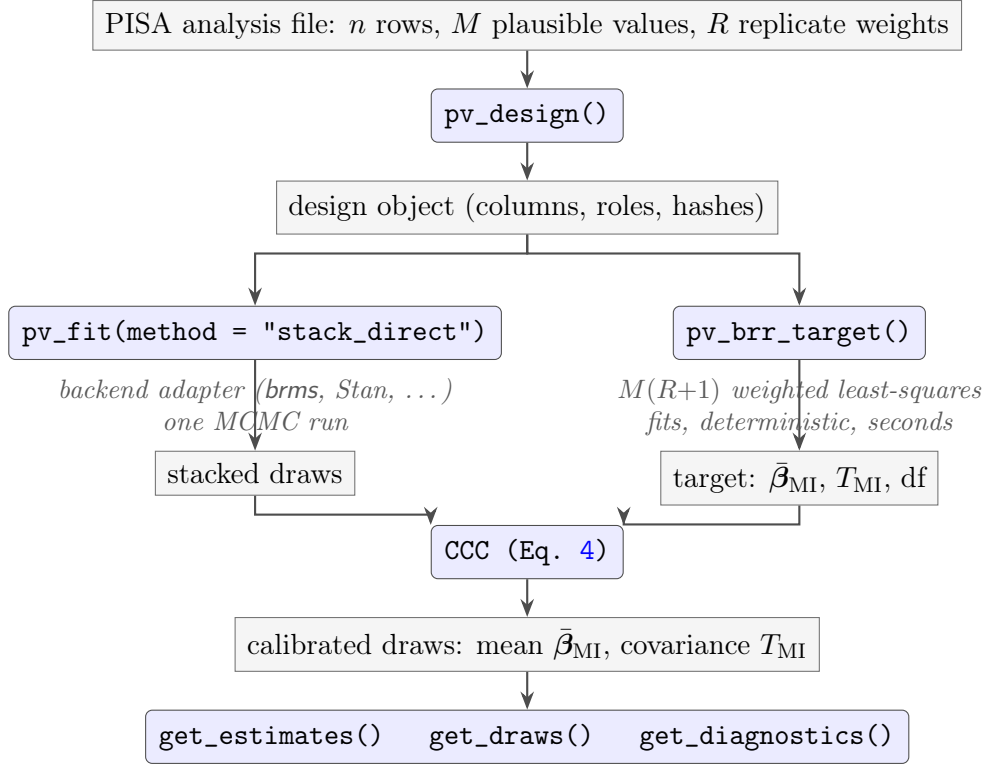


FIGURE 1. The `pvstackr` workflow. The design stage declares the data roles; the fit and target stages then run separately, one sampler run on the stacked data and $M(R+1)$ deterministic weighted least-squares fits; the Cholesky calibration correction maps the stacked draws to the target, and the accessors return estimates, draws, and diagnostics.

The fit stage enforces its preconditions rather than attempting repairs. Formulas containing random-effect terms are rejected, because the target engine and the coverage argument cover fixed effects only. Explicit priors on the coefficients trigger a warning status, because the target is a flat-prior least-squares quantity and a strongly informative prior would separate the stacked center from the target center. That separation is itself monitored: the calibration step compares the stacked posterior mean with the target center, relative to target scale, and issues a warning at 10^{-2} and a blocked fit at 5×10^{-2} . A blocked fit returns diagnostics but no estimates. In practice the warning threshold is reached when the sampler provides too few draws for weakly identified coefficients; the application below sets the draw budget so that the diagnostic remains well inside the acceptable range, and the supplement shows the blocking threshold reached under an informative prior.

3.3. Installation and the minimal call. The package installs from GitHub (Lee, 2026), and its documentation site provides ten vignettes in two tracks, five applied and five on the method.

```
# install.packages("remotes")
remotes::install_github("joonho112/pvstackr@v0.2.0")
library(pvstackr)
```

The minimal reportable analysis consists of three calls: the design is declared, the target is assembled, and the stacked fit is run and calibrated.

```
design <- pv_design(dat, OUTCOME ~ x1 + x2, pv_suffix = "READ",
                  weight_col = "W_FSTUWT")
target <- pv_brr_target(dat, OUTCOME ~ x1 + x2,
                      pv_cols      = design$pv_cols,
                      weight_col   = design$weight_col,
                      rep_weight_cols = design$rep_weight_cols,
                      fay_k        = design$fay_k,
                      df_method    = "barnard_rubin",
                      df_complete  = 80)
fit <- pv_fit(dat, OUTCOME ~ x1 + x2, target = target,
             method = "stack_direct",
             control = pv_control(method = "stack_direct",
                                  backend = "brms"))
```

OUTCOME is a placeholder that the package replaces with each plausible-value column in turn. In the next section this pattern is applied to real data.

4. Applications

In this section the workflow is applied to reading proficiency in the PISA 2022 American sample. Every number and console display below is produced by a seeded, publicly available replication archive (see the Software and reproducibility statement), and the complete script is listed in the supplement.

4.1. Data and model. The analysis file starts from the public-use PISA 2022 student and school questionnaire files (OECD, 2023). The released United States sample contains 4,552 students in 154 schools. Listwise deletion on the model variables, socioeconomic status (the PISA index of economic, social and cultural status, ESCS), gender, immigrant background, and the school questionnaire's community-size item, leaves 3,915 students in 142 schools; the supplement reports the step-by-step cascade.

Each retained record carries the ten reading plausible values PV1READ to PV10READ, the final student weight W_FSTUWT, and the 80 BRR-Fay replicate weights W_FSTURWT1 to W_FSTURWT80.

The substantive analysis is the classic within/between decomposition (Allison, 2009; Rabe-Hesketh & Skrondal, 2022); the question is whether a student characteristic predicts reading within a school at the same rate at which its school-level composition predicts school performance. Each student-level predictor x_{ij} is therefore split into a weighted school mean $\bar{x}_{.j}$ and a within-school deviation $x_{ij} - \bar{x}_{.j}$, and the two parts enter a two-level random-intercept model as separate regressors, together with two school-level covariates from the sampling frame and the school questionnaire,

$$\begin{aligned} y_{ij}^{(m)} = & \beta_0 + \beta_1^W (\text{escs}_{ij} - \overline{\text{escs}_{.j}}) + \beta_2^W (\text{female}_{ij} - \overline{\text{female}_{.j}}) \\ & + \beta_3^W (\text{immig}_{ij} - \overline{\text{immig}_{.j}}) + \beta_1^B \overline{\text{escs}_{.j}} + \beta_2^B \overline{\text{female}_{.j}} \\ & + \beta_3^B \overline{\text{immig}_{.j}} + \gamma_1 \text{private}_j + \gamma_2 \text{city}_j + u_j + \varepsilon_{ij}^{(m)}, \end{aligned} \quad (5)$$

where $u_j \sim N(0, \tau^2)$ is a school-level random intercept and $\varepsilon_{ij}^{(m)} \sim N(0, \sigma^2)$ is a student-level residual. The estimands are the fixed effects: the within-school gradients β^W , the between-school gradients β^B , the school-sector and location contrasts, and the differences $\beta^B - \beta^W$ (Rabe-Hesketh & Skrondal, 2022, ch. 3).

A word on how the package estimates model (5) is in order, because the random intercept, which represents unobserved heterogeneity between schools and motivates the decomposition itself, is not estimated on the route the package takes. Provided u_j is uncorrelated with the covariates, model (5) implies that the marginal mean of $y_{ij}^{(m)}$ is the fixed-effect part alone, and the design-weighted least-squares estimator of that mean function, applied once per plausible value, is consistent for the same fixed effects. The within-school dependence that u_j induces is instead absorbed by the variance estimation: the BRR replicates perturb whole schools, so the covariance of Equation (2) reflects the clustering without a random effect being fitted, much as Rabe-Hesketh and Skrondal (2006) account for PSUs through the sandwich estimator rather than through an additional level in the model. Two considerations motivate this route. First, the public-use files release replicate weights at the student level only, so a replicate-based design variance for the variance components (τ^2, σ^2) is not identified from public data (Lee et al., 2026). Second, weighted least squares forgoes some efficiency relative to weighted pseudo-maximum-likelihood estimation of model (5) (Rabe-Hesketh & Skrondal, 2006); the engine developed for the companion

paper fits the weighted random-intercept model directly and calibrates its fixed-effect block, and porting that path into the package interface is the first item of planned development. Two features of the sample are also worth noting in advance: the sampling frame identifies six private schools among the 154 released, of which four survive the analytic exclusions, so the private-school contrast rests on four first-stage units; and the school-level gradients in general rest on far fewer effective units than the 3,915 students might suggest.

4.2. Declaring the design and assembling the target. The design stage records which columns play which role, and resolves the plausible-value and replicate-weight sets by name.

```
design <- pv_design(dat, OUTCOME ~ escs_dev + female_dev + immig_dev +
                    escs_mn + female_mn + immig_mn + private + city,
                    pv_suffix = "READ", expected_M = 10L, expected_R = 80L,
                    weight_col = "W_FSTUWT", id_cols = c("CNTSCHID",
                    "CNTSTUID"))
design

pvstackr design
rows: 3915
formula: OUTCOME ~ escs_dev + female_dev + immig_dev + escs_mn + female_mn +
immig_mn + private + city
plausible values: 10
final weight: W_FSTUWT
replicate weights: 80
fay_k: 0.5
design hash: 90d5af6c
```

The target stage runs the $M(R + 1) = 810$ weighted least-squares fits of Equations (1) and (2) and returns the assembled center, covariance, and degrees of freedom. The complete-data degrees of freedom for the Barnard and Rubin (1999) correction are set to 80, the number of variance strata behind the replicate weights (Wolter, 2007); the supplement shows that the classic degrees-of-freedom choice moves no interval half-width by more than two percent.

```
target <- pv_brr_target(dat, formula = design$formula,
                        pv_cols      = design$pv_cols,
                        weight_col    = design$weight_col,
                        rep_weight_cols = design$rep_weight_cols,
```

```

      fay_k      = design$fay_k,
      id_cols    = design$id_cols,
      df_method  = "barnard_rubin",
      df_complete = 80)

target

```

```

|pvstackr BRR-Fay target
|  fixed effects: 9
|  plausible values: 10
|  replicate weights: 80
|  fay_k: 0.5
|  df method: barnard_rubin
|  interval role: coverage_barnard_rubin
|  source: external_brr_fay_rubin

```

On a MacBook Pro with an Apple M4 Max processor, the 810 fits complete in 0.5 seconds. The supplement verifies the assembled target against an independent implementation, `svrepdesign()` and `svyglm()` from the `survey` package (Lumley, 2004) with Fay’s method and the same combining arithmetic; point estimates and standard errors agree to fourteen significant digits.

4.3. One fit and the calibrated posterior. The fit stage requires a sampler, and with `brms` and `cmdstanr` installed no adapter code is needed: the bundled backend is selected through the control object, and a single call runs the MCMC fit and calibrates it. The stacked model is a weighted Gaussian regression on the prepared long data, with the plausible-value rows entering the likelihood at weight $\tilde{w}_i/M$ as in Equation (3). Flat default priors on the coefficients keep the stacked center aligned with the least-squares target; the package flags any explicitly supplied prior with a warning status.

```

ctrl <- pv_control(method = "stack_direct", backend = "brms",
                  chains = 4, iter = 5000, warmup = 1000, cores = 4,
                  seed = 20260711)
fit <- pv_fit(dat, formula = design$formula, target = target,
             method = "stack_direct", control = ctrl)

fit

```

```

|pvstackr fit
|  method: stack_direct

```

```

status: ok
fixed effects: 9
target: external_brr_fay_rubin
draws: 16000 x 9
diagnostics: preflight, sampler, sampler_gate, stack_fit, stack_fit_warnings,
             ccc

```

The stacked data set has $3,915 \times 10 = 39,150$ rows, and the four chains deliver 16,000 post-warmup draws in about three minutes. Since there is a single fit, there is a single set of convergence diagnostics: the maximum split- $\hat{R}$ is 1.001, the minimum bulk effective sample size is 7,494, and there are no divergent transitions. The calibration step reports a maximum center separation of 0.006 relative to target scale, well inside the acceptable range. `get_estimates(fit)` returns the reportable table; the excerpt below shows the metadata columns, and Table 3 reports all nine coefficients.

```

est <- get_estimates(fit)
est[2:4, c("term", "estimate", "se", "df", "conf_low", "conf_high",
           "coverage_claim_allowed")]

```

	term	estimate	se	df	conf_low	conf_high
2	b_escs_dev	25.346354	2.309663	56.41494	20.72029	29.97241
3	b_female_dev	21.974729	3.452741	73.99769	15.09499	28.85447
4	b_immig_dev	-1.662613	6.049544	57.62185	-13.77378	10.44856
	coverage_claim_allowed					
2		TRUE				
3		TRUE				
4		TRUE				

The estimates are in accordance with previous research on reading achievement in the United States. Within schools, a one-unit difference in ESCS is associated with a difference of 25.3 points in reading performance, and girls score 22.0 points higher than boys, controlling for the other covariates; the within-school immigrant gap is indistinguishable from zero once ESCS is held fixed. The between-school gradients are considerably steeper than their within-school counterparts. Note that the private-school contrast rests on four first-stage units, and its 95% interval accordingly spans 98 points.

TABLE 3. Calibrated-stack estimates for the PISA 2022 United States reading model

Term	Estimate	SE	df	95% CI	FMI
Intercept	470.58	23.48	74.3	[423.80, 517.37]	0.04
ESCS (within school)	25.35	2.31	56.4	[20.72, 29.97]	0.15
Female (within school)	21.97	3.45	74.0	[15.09, 28.85]	0.04
Immigrant background (within school)	−1.66	6.05	57.6	[−13.77, 10.45]	0.15
School mean ESCS	80.73	7.36	71.6	[66.05, 95.41]	0.06
School share female	36.16	46.89	74.4	[−57.27, 129.59]	0.04
School share immigrant	60.43	22.52	76.3	[15.57, 105.29]	0.02
Private school	−10.00	24.58	76.2	[−58.94, 38.95]	0.02
City location	−2.07	7.45	76.1	[−16.92, 12.77]	0.02

Note. Estimates and standard errors equal the BRR-Fay/Rubin target by construction; degrees of freedom are Barnard–Rubin with complete-data degrees of freedom 80; FMI is the fraction of missing information. All rows carry `interval_role = "coverage_barnard_rubin"` and `coverage_claim_allowed = TRUE`. $n = 3915$ students, 142 schools, $M = 10$, $R = 80$.

4.4. Probability statements from the calibrated draws. The quantities reported up to this point could equally have been produced by a design-based frequentist package. The calibrated fit additionally provides the posterior draws behind them: `get_draws(fit)` returns the 16,000 calibrated draws, whose mean and covariance equal the target exactly, so derived quantities inherit design-based uncertainty without delta-method approximations. We consider the difference between the between-school and within-school gradients, which has attracted considerable interest as a contextual or compositional effect of socioeconomic status (Rabe-Hesketh & Skrondal, 2022). The difference is computed from the draws in a single line:

```
d <- get_draws(fit)
dif <- d[, "b_escs_mn"] - d[, "b_escs_dev"]
c(mean = mean(dif), sd = sd(dif), pr_gt0 = mean(dif > 0))
```

	mean	sd	pr_gt0
	55.3863	7.8319	1.0000

For ESCS the difference is 55.4 points (95% interval 40.1 to 70.8), and it is positive in all 16,000 calibrated draws; for immigrant background it is 62.1 points with $\Pr(\Delta > 0) = 0.996$. Figure 2a displays both densities. Probability summaries of

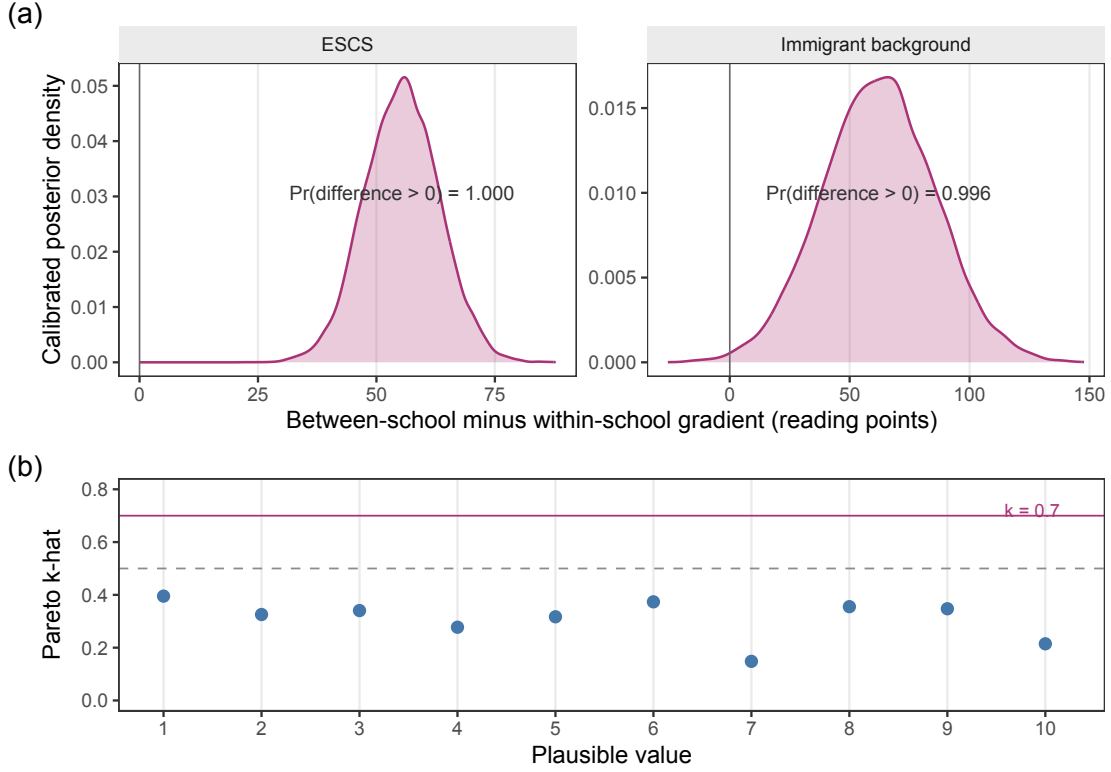


FIGURE 2. Calibrated posteriors of the between–within differences and the PSIS diagnostic. (a) Calibrated posterior densities of the difference between the between-school and within-school gradients for ESCS and immigrant background, with posterior exceedance probabilities. (b) Pareto $\hat{k}$ for the importance reweighting of the stacked draws toward each per-PV posterior; all ten values lie below the 0.7 threshold.

this kind are what Kaplan and Harra (2024) recommend reporting to policy audiences, here attached to design-based rather than model-based uncertainty. Their substantive interpretation, however, requires care (Rabe-Hesketh & Skrondal, 2022, ch. 3). If school-mean ESCS is uncorrelated with omitted school-level determinants of achievement, the difference is a contextual effect, an association of the school’s socioeconomic composition with achievement over and above the student’s own status. If that exogeneity assumption fails, the same difference reflects school-level confounding: students of high socioeconomic status may be concentrated in schools that would achieve well regardless of their composition, so that the school mean is correlated with the school-level error. The posterior places essentially all of its mass on a positive difference, but it cannot distinguish between these explanations.

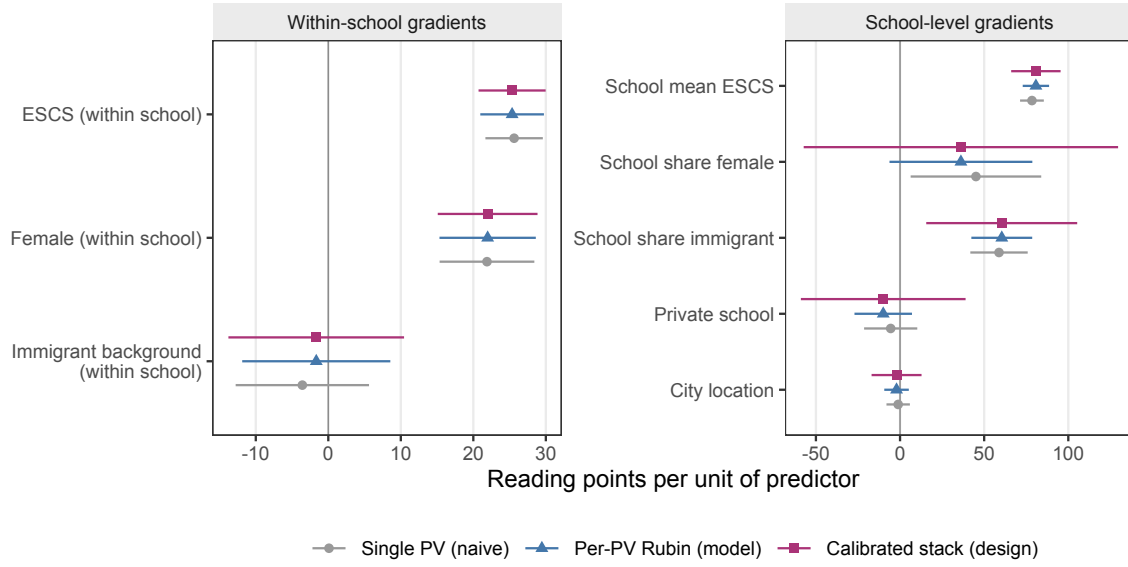


FIGURE 3. Estimates and 95% intervals under three inferential routes. Point estimates agree across the three routes, whereas the interval widths differ. The naive single-PV route understates uncertainty for the measurement-sensitive within-school gradients, and the model-based Rubin route understates it for the school-level gradients, where the design-based intervals are roughly twice as wide.

4.5. Agreement with the per-PV workflow and the anatomy of the standard errors. The conventional route fits the model once per plausible value and pools the results with Equation (1). Running the same `brms` specification ten times and passing the draw matrices to `pv_fit_reference()` gives this model-based reference; `pv_compare_methods()` aligns the finished fits, and Figure 3 overlays the three routes. Point estimates agree to within 0.04 reading points across all nine coefficients, with a maximum standardized difference of 0.002, which is the empirical counterpart of the stacked point identity. The diagnostic path `pv_fit_stack_psis()` reweights the stacked draws toward each per-PV posterior; all ten Pareto $\hat{k}$ values fall between 0.15 and 0.40 (Figure 2b), and its pooled standard errors reproduce the per-PV reference within two percent, which confirms that the single stacked fit is an adequate proposal for every per-PV posterior in this application.

Table 4 and Figure 4 decompose the sources of uncertainty. Moving from the naive single-PV analysis to per-PV Rubin pooling adds the measurement component; the increase tracks the fraction of missing information, and it is visible for the ESCS and immigrant within-school gradients (FMI of about 0.15) but negligible where the

TABLE 4. Standard errors under three inferential routes

Term	Estimate	Standard error			Ratio
		Naive PV1	Per-PV Rubin	Calibrated stack	
Intercept	470.58	10.06	11.01	23.48	2.13
ESCS (within school)	25.35	2.02	2.23	2.31	1.04
Female (within school)	21.97	3.33	3.39	3.45	1.02
Immigrant background (within school)	-1.66	4.69	5.18	6.05	1.17
School mean ESCS	80.73	3.60	3.98	7.36	1.85
School share female	36.16	19.80	21.57	46.89	2.17
School share immigrant	60.43	8.70	9.21	22.52	2.45
Private school	-10.00	8.06	8.69	24.58	2.83
City location	-2.07	3.55	3.71	7.45	2.01

Note. Estimate is the calibrated-stack point estimate; the per-PV Rubin estimates agree with it to within 0.04 points, while the naive PV1 point estimate depends on which plausible value is kept (see text). Naive PV1 is the model-based posterior standard deviation from a single-plausible-value fit; Per-PV Rubin pools ten model-based fits by Equation (1); Calibrated stack is the design-based BRR-Fay/Rubin standard error. Ratio is Calibrated stack over Per-PV Rubin.

FMI is 0.02. Moving from model-based to design-based variance then adds the sample design, and the increase is concentrated where the effective number of first-stage units is small: the within-school gradients grow by 2 to 17 percent, whereas the school-level gradients grow by factors of 1.85 to 2.83. A model-based Bayesian analysis of these data would therefore report intervals for the school-level gradients roughly half as wide as the design-based ones, however satisfactory its convergence diagnostics may be. The single-PV shortcut errs in the other direction, and its point estimate additionally depends on which plausible value happens to be kept; the within-school immigrant coefficient, for instance, is -3.6 under PV1 and -1.7 pooled.

4.6. Cost. The full accounting for this application, on the same MacBook Pro (Apple M4 Max) with four parallel chains, is as follows: the target stage, 810 weighted least-squares fits, takes 0.5 seconds; the stacked fit with calibration, 177 seconds; the ten-fit per-PV reference, 191 seconds; and the PSIS diagnostics, 3 seconds. The stacked fit is thus not faster than the per-PV route at equal sampler settings, and we make no such claim; its advantage is one convergence assessment instead of ten and a single model object to archive. The difference appears in the design-based uncertainty: a fully Bayesian BRR analysis run by brute force, with one MCMC fit per plausible value and replicate weight, would require 810 sampler runs, over four hours at the per-fit times observed here, to produce the covariance that the target stage assembles in 0.5 seconds.

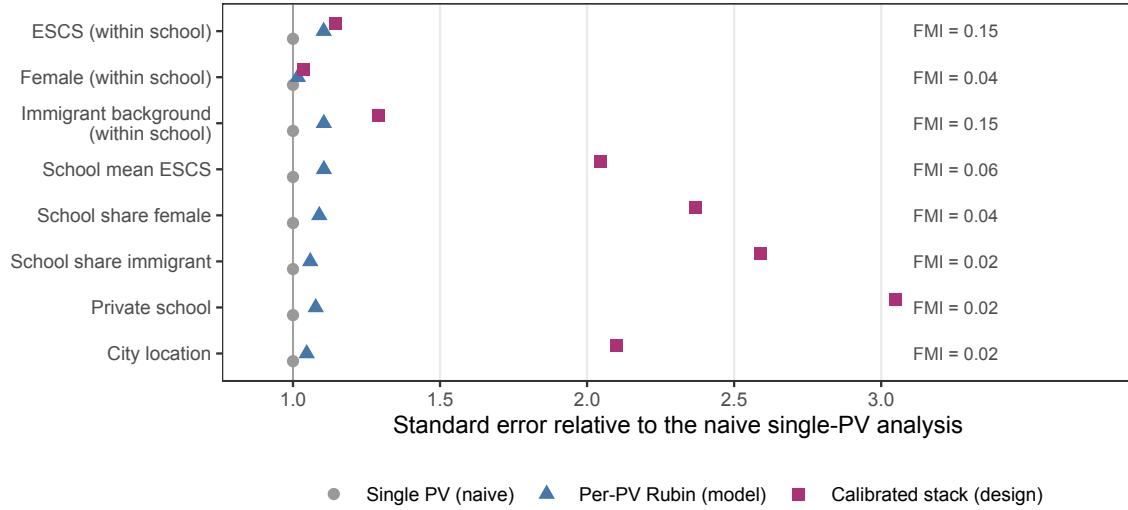


FIGURE 4. Anatomy of the standard errors relative to the naive single-PV analysis. Per-PV Rubin pooling adds the measurement component, which tracks the fraction of missing information; the calibrated stack adds the design component, which concentrates in the school-level gradients resting on 142 first-stage units.

4.7. Planned development. Version 0.2.0 covers fixed effects of survey-weighted linear models with BRR-Fay replicate weights; the bundled `brms` backend was added in this version. Two extensions are planned. The first is the weighted random-intercept path: the engine developed for the companion paper fits model (5) by weighted pseudo-maximum-likelihood (cf. Rabe-Hesketh & Skrondal, 2006) and calibrates its fixed-effect block, and porting that path into the package interface would allow the same workflow to report variance components; a design-based replicate target for the variance components remains an open problem, because the public files identify replicate weights at the student level only (Lee et al., 2026). The second is jackknife replication, which would extend the target stage to the IEA studies such as TIMSS and PIRLS (Zheng et al., 2026).

5. Conclusions

We have described an R package that connects two computations which existing software keeps separate. The sampler produces a joint posterior sample for the fixed effects; a deterministic target stage obtains the center, the covariance, and the degrees of freedom of the reported inference from the replicate weights; and an exact affine transformation connects the two. For the applied analyst the practical consequences

are threefold. First, a Bayesian plausible-value analysis requires one MCMC run and one convergence assessment rather than M . Second, the reported uncertainty coincides with the design-based BRR-Fay/Rubin quantity that the frequentist packages and the official reporting conventions produce, to machine precision in our verification against the **survey** package. Finally, the posterior draws are retained through the calibration, so probability statements of the kind advocated by Kaplan and Harra (2024) can be reported with design-based rather than model-based uncertainty.

The scope of the package is restricted to the fixed effects of weighted linear models. The construction behind every interval is recorded, and exactly one construction is treated as coverage-claimable; as noted above, we make no efficiency claim for the stacked fit itself. Within the Bayesian workflow that Kaplan and Harra (2024) propose for large-scale assessments, the package automates the step their case study left open, the connection between posterior summaries and the replicate weights, and it extends to the Bayesian case the software support for replication methods that Zheng et al. (2026) call for. For an analyst who already fits weighted regression models with **brms**, the bundled backend makes this workflow available without further programming.

List of abbreviations

BRR: balanced repeated replication; CCC: Cholesky calibration correction; CI: credible/confidence interval; df: degrees of freedom; ESCS: PISA index of economic, social and cultural status; FMI: fraction of missing information; IEA: International Association for the Evaluation of Educational Achievement; MCMC: Markov chain Monte Carlo; MI: multiple imputation; NAEP: National Assessment of Educational Progress; OECD: Organisation for Economic Co-operation and Development; PIRLS: Progress in International Reading Literacy Study; PISA: Programme for International Student Assessment; PSIS: Pareto-smoothed importance sampling; PV: plausible value; SE: standard error; TIMSS: Trends in International Mathematics and Science Study.

Software and reproducibility. The workflow is implemented in the open-source R package **pvstackr** (project home page <https://github.com/joonho112/pvstackr>; documentation at <https://joonho112.github.io/pvstackr/>), released under the MIT license and requiring R (≥ 4.3); the core package has no hard dependencies beyond base R, and the bundled sampling backend uses **brms** and **cmdstanr**. A public replication package for this article is available at <https://github.com/joonho112/pvstackr-tutorial->

replication and contains all analysis code and derived exhibits but redistributes no OECD unit-record data.

Data availability. The PISA 2022 data are publicly available from the OECD (OECD, 2023) at <https://www.oecd.org/pisa/data/>. Under the current PISA-specific access terms the unit-record files may not be redistributed by a third party, so the replication package supplies the data-build scripts rather than the student and school records themselves.

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Online Supplement

pvstackr: An R Package for Bayesian Plausible-Value Analysis
with One Stacked Fit Calibrated to Replicate-Weight Variance Targets

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Contents

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Appendix A. Sample construction and descriptive statistics

The analysis file derives from the OECD PISA 2022 public-use student and school questionnaire files. Table A.1 reports the analytic-sample cascade. Student-level exclusions use the model variables directly; the school-level exclusion removes the eight schools for which the community-size item (SC001Q01TA) is missing, after which the frame-based sector variable (PRIVATESCH) is complete for all remaining schools.

TABLE A.1. Analytic-sample cascade, PISA 2022 United States

Step	Students	Schools
Released sample	4552	154
Drop missing ESCS	4342	150
Drop missing gender	4337	150
Drop missing immigrant background	4135	150
Drop schools with missing community size	3915	142

Variable codings follow the PISA 2022 codebook: `female` = 1 if ST004D01T = 1; `immig` = 1 if IMMIG $\in \{2, 3\}$ (second- or first-generation); `private` = 1 if the sampling-frame variable PRIVATESCH equals "private"; `city` = 1 if SC001Q01TA ≥ 4

(community of 100,000 or more). School means are weighted by the final student weight within school; deviations subtract the school mean from the student value. Table A.2 reports weighted descriptive statistics on the analytic sample.

TABLE A.2. Weighted descriptive statistics, analytic sample ($n = 3915$)

Variable	Mean	SD	Min	Max
ESCS	0.085	0.970	-3.37	3.53
Female	0.506	0.500	0.00	1.00
Immigrant background	0.236	0.425	0.00	1.00
School mean ESCS	0.085	0.505	-1.34	1.21
School share female	0.506	0.085	0.18	1.00
School share immigrant	0.236	0.221	0.00	1.00
Private school	0.047	0.212	0.00	1.00
City location	0.438	0.496	0.00	1.00
Reading, first plausible value	509.79	111.35	162.63	847.22

Note. Means and standard deviations use the final student weight W_FSTUWT. Four of the 142 schools are private; 65 are in communities of 100,000 or more.

Appendix B. Complete replication path

The replication archive, publicly available at <https://github.com/joonho112/pvstackr-tutorial-replication>, contains a staged pipeline (code/01_build_data.R through code/11_osm_budget_demo.R, driven by run_all.R) that produces every number, table, figure, and console display in the manuscript and in this supplement, together with a condensed linear script, `replication-script.R`, that reproduces the core analysis in about seven minutes. The staged pipeline writes verbatim console captures to `output/txt/`, and the console displays in the manuscript reproduce those files verbatim. The listing below reproduces the core of the linear script.

```
library(pvstackr)
PV_COLS <- paste0("PV", 1:10, "READ")
RHS <- "escs_dev + female_dev + immig_dev + escs_mn + female_mn +
      immig_mn + private + city"
F <- as.formula(paste("OUTCOME ~", RHS))

dat <- readRDS("output/rds/pisa22_usa_read_hybrid.rds")

design <- pv_design(dat, F, pv_suffix = "READ", expected_M = 10L,
```

```

      expected_R = 80L, weight_col = "W_FSTUWT",
      id_cols = c("CNTSCHID", "CNTSTUID"))
target <- pv_brr_target(dat, F, pv_cols = design$pv_cols,
      weight_col = design$weight_col,
      rep_weight_cols = design$rep_weight_cols,
      fay_k = design$fay_k, id_cols = design$id_cols,
      df_method = "barnard_rubin", df_complete = 80)

ctrl <- pv_control(method = "stack_direct", backend = "brms",
      chains = 4, iter = 5000, warmup = 1000, cores = 4,
      seed = 20260711)
fit <- pv_fit(dat, F, target = target, method = "stack_direct",
      control = ctrl)
get_estimates(fit)

d <- get_draws(fit)
ctx <- d[, "b_escs_mn"] - d[, "b_escs_dev"]
c(mean = mean(ctx), sd = sd(ctx), pr_gt0 = mean(ctx > 0))

```

The per-plausible-value reference, the PSIS diagnostic path, and the method comparison follow in `replication-script.R` steps 5 to 7 and in the staged scripts `code/04_fit_perpv.R`, `code/05_fit_psis.R`, and `code/06_compare.R`.

The injected-adapter mechanism. Any other Bayesian engine is attached to the stacked fit through three injected functions: one that fits the prepared stacked data using the ready-made formula `.pvstackr_y | weights(.pvstackr_weight) ~ ...`, one that returns the draws as a plain base matrix, and one that reports the sampler diagnostics. All three are needed for a reportable fit; an adapter that supplies only the first two leaves the sampler evidence incomplete and the fit is blocked.

The bundled backend is built from exactly these three functions, and the package exports them, so the injected route can reuse the bundled implementation and replace only the piece that differs. Injecting all three runs the same code the bundled backend runs and yields the same estimates; what differs is the recorded provenance, which marks the fit as injected, and the fact that an injected adapter is responsible for its own cache directory.

```

fit <- pv_fit(dat, F, target = target, method = "stack_direct",
      control = pv_control(method = "stack_direct",
        backend = "cmdstanr"),
      fit_function = pv_backend_brms_fit_function,

```

```

draws_function      = pv_backend_brms_draws_function,
diagnose_function   = pv_backend_brms_sampler_diagnostics,
cache_dir           = NULL)

```

To attach a different engine, replace the members that differ and keep the rest. The fit function receives `formula`, `data`, `family`, `prior`, `chains`, `iter`, `warmup`, `cores`, `seed`, `backend`, `file`, and `file_refit`; the draws function receives the fitted object and returns a plain base matrix whose columns are the fixed effects and the residual scale; the diagnose function receives the fitted object and returns the sampler summary.

Appendix C. Independent verification of the BRR-Fay target

The target assembled by `pv_brr_target()` was recomputed with an independent implementation: a replicate-weight design declared through `survey::svrepdesign()` with `type = "Fay"`, `rho = 0.5`, and `mse = TRUE`, one `survey::svyglm()` fit per plausible value, and the combining rules applied to the resulting coefficient vectors and covariance matrices. Table C.1 reports both sets of standard errors. The largest relative difference across the nine coefficients is 1.4×10^{-14} for the point estimates and 1.7×10^{-14} for the standard errors, that is, agreement to machine precision. Note that the `mse = TRUE` option is required for exact agreement: it centers each replicate deviation at the full-sample estimate, which is the convention that `pvstackr` implements and that the OECD documentation prescribes.

TABLE C.1. Cross-check of the target against the `survey` package

Term	Estimate	SE (pvstackr)	SE (survey)	Rel. diff. SE
Intercept	470.584	23.482	23.482	7.6×10^{-16}
ESCS (within school)	25.346	2.310	2.310	1.7×10^{-14}
Female (within school)	21.975	3.453	3.453	2.6×10^{-16}
Immigrant background (within school)	-1.663	6.050	6.050	4.4×10^{-16}
School mean ESCS	80.733	7.364	7.364	2.4×10^{-15}
School share female	36.162	46.894	46.894	9.1×10^{-16}
School share immigrant	60.431	22.524	22.524	0
Private school	-9.995	24.575	24.575	1.4×10^{-16}
City location	-2.075	7.454	7.454	1.8×10^{-15}

Appendix D. Degrees-of-freedom sensitivity

The manuscript’s intervals use the Barnard–Rubin degrees of freedom with complete-data degrees of freedom set to 80, the number of variance strata behind the replicate weights. Table D.1 compares the resulting 95% interval half-widths with those under the classic Rubin degrees of freedom, which ignore the complete-data limit and are much larger here because the fractions of missing information are modest. The Barnard–Rubin choice widens every interval, by 1.6 to 1.9 percent; conclusions are unaffected, and the package labels only the Barnard–Rubin construction coverage-claimable.

TABLE D.1. Interval half-widths under two degrees-of-freedom conventions

Term	SE	df (classic)	df (BR)	Half-width ratio
Intercept	23.482	6558.2	74.3	1.016
ESCS (within school)	2.310	384.2	56.4	1.019
Female (within school)	3.453	5702.1	74.0	1.016
Immigrant background (within school)	6.050	423.6	57.6	1.019
School mean ESCS	7.364	2684.5	71.6	1.017
School share female	46.894	6630.8	74.4	1.016
School share immigrant	22.524	23549.4	76.3	1.016
Private school	24.575	21963.7	76.2	1.016
City location	7.454	19668.8	76.1	1.016

Note. BR = Barnard–Rubin with complete-data degrees of freedom 80. Half-width ratio is the BR interval half-width over the classic one; standard errors are identical by construction.

Appendix E. School-mean construction sensitivity

The manuscript computes school means with the final student weight. Recomputing the entire target with unweighted school means alters no substantive conclusion: the within-school gradients change by at most 0.5 reading points, the school-level gradients by at most 2.4 points against design-based standard errors an order of magnitude larger, and the intercept by 1.6 points. Table E.1 reports the full comparison.

Appendix F. Extended diagnostics and a demonstration of the safeguards

MCMC diagnostics of the single stacked fit. The four chains, of 5,000 iterations each with 1,000 warmup, deliver 16,000 retained draws; the maximum split- $\hat{R}$ is 1.001, the

TABLE E.1. Weighted versus unweighted school means

Term	Weighted means		Unweighted means	
	Estimate	SE	Estimate	SE
Intercept	470.584	23.482	472.158	20.118
ESCS (within school)	25.346	2.310	25.373	2.309
Female (within school)	21.975	3.453	22.037	3.424
Immigrant background (within school)	-1.663	6.050	-1.208	6.049
School mean ESCS	80.733	7.364	80.755	7.267
School share female	36.162	46.894	33.808	40.525
School share immigrant	60.431	22.524	59.814	22.670
Private school	-9.995	24.575	-9.472	24.951
City location	-2.075	7.454	-1.888	7.439

minimum bulk effective sample size is 7,494, the minimum tail effective sample size is 9,980, and there are no divergent transitions.

Calibration diagnostics. The Cholesky calibration reports a root-mean-square relative center separation of 0.004 and a maximum of 0.006 across the nine coefficients, well inside the acceptable range (warning threshold 10^{-2} , blocking threshold 5×10^{-2}). With the package’s default draw budget of 2000 iterations per chain, an archived companion run (`code/11_osm_budget_demo.R`) reproduces a maximum separation of 0.0103, which exceeds the warning threshold for the weakly identified school-share coefficients; raising the budget to 5000 iterations restores an acceptable separation. This is the intended behavior of the diagnostic: because the calibration fixes the mean of the draws at the target, Monte Carlo noise in the stacked center must be detected before the calibration is applied.

PSIS diagnostics. Pareto $\hat{k}$ for the importance reweighting of the stacked draws toward the ten per-PV posteriors: 0.395, 0.326, 0.341, 0.277, 0.317, 0.374, 0.148, 0.355, 0.347, 0.215, all below 0.5 and well below the 0.7 threshold at which the path withholds its estimates. The pooled `stack_psis` standard errors agree with the per-PV reference within two percent on every coefficient (maximum absolute log standard-error ratio 0.015), and the pooled estimates agree with the reference to a maximum absolute standardized difference of 0.011.

A demonstration of the safeguards: explicit priors. Refitting the stacked model with an explicit coefficient prior, `normal(0, 25)` on all slopes, exercises two safeguards at once. The prior policy is triggered first: the external target is a flat-prior least-squares quantity, so any explicit prior is flagged. The center-separation check then

escalates the warning to a blocked fit. A normal prior with standard deviation 25 is far from flat for the weakly identified school-level coefficients in this model (the school-mean ESCS gradient is 81 points, and the school-share-female gradient has a stacked posterior spread of about 20 points), and the prior shrinkage moves the stacked center 0.379 target standard deviations away from the target at the most affected coefficient, well past the blocking threshold of 0.05. The fit returns diagnostics and reason codes but no estimates.

```
fit_prior <- pv_fit(dat, F, target = target, method = "stack_direct",
                  control = ctrl,
                  prior = brms::set_prior("normal(0, 25)", class = "b"))
fit_prior
```

```
|pvstackr fit
|  method: stack_direct
|  status: blocked
|  fixed effects: 0
|  target: external_brr_fay_rubin
|  draws: not retained
|  diagnostics: preflight, sampler, sampler_gate, ccc, redaction
|  reason codes: explicit_prior_warning, center_separation_red
|  warnings: 2
```

A blocked fit keeps only the evidence for the decision. The reportable estimates and the draws are withheld, and the retained diagnostics are the preflight record, the sampler summary and its gate, the center-separation gate under `ccc`, and a record of what was redacted.

This is the intended failure mode: the calibration recenters draws, and recentering across a gap of this size would misrepresent what the sampler explored. An analyst who requires informative priors for substantive reasons should use the per-PV route and report the resulting intervals as model-based.

Appendix G. Environment, seeds, and timing

All results were produced on a MacBook Pro with an Apple M4 Max processor (16 cores, macOS), with R 4.6.0, `pvstackr` 0.2.0, `brms` 2.23.0, `cmdstanr` 0.8.0 with CmdStan 2.38.0, `posterior` 1.7.0, `loo` 2.10.1, and `survey` 4.5. The stacked fit uses seed 20260711; per-PV fit m uses seed $20260711 + m$. Wall-clock times for the staged pipeline: data construction 0.5 s; BRR-Fay target (810 weighted least-squares fits) 0.5 s; stacked fit

plus calibration 177.1 s; ten per-PV fits 190.9 s (mean 19.1 s); PSIS log-ratios and diagnostics 3.2 s; **survey** cross-check 1.1 s.

The analyses require **pvstackr** version 0.2.0 or later: version 0.2.0 adds the bundled **brms** backend used throughout, and version 0.1.1 corrects the PSIS-weighted covariance computation used by the **stack_psis** diagnostic path, whose standard errors are inflated under version 0.1.0 (**stack_direct** and **per_pv** point estimates and standard errors are unaffected).